

B.Tech Degree V Semester (Supplementary) Examination June 2011

ME 503 MECHANICS OF MACHINERY (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

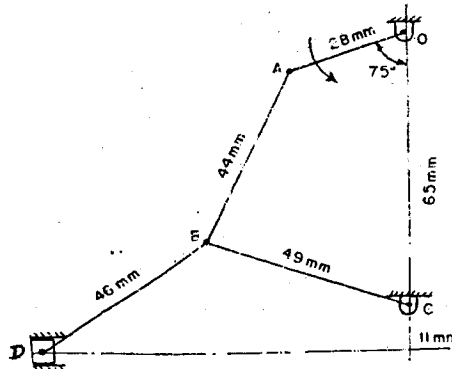
PART A (Answer all questions)

(8x5=40)

- I. a) Sketch and describe the working of two different types of Quick Return Motion Mechanisms.
 b) State and explain Kennedy's theorem for instantaneous centres.
 c) Explain the overlay method in the synthesis of mechanism.
 d) Explain what is meant by inverse kinematic transformations.
 e) Derive an expression for the velocity of sliding between a pair of involute gear teeth.
 f) State the advantages of involute profile over cycloidal profile gear teeth.
 g) Derive expressions for the displacement, velocity and acceleration for a roller follower, when the roller is on the flank of a tangent cam.
 h) Derive an expression for the efficiency of a screw jack.

PART B

- II. In figure, the crank OA is rotating at a uniform speed of 500rpm in the anticlockwise direction. Determine the linear velocity of slider D and the angular velocity of the link BD, when the crank is inclined at an angle of 75° to the vertical. The dimensions of various links are BD=46mm, OA=28mm, AB=44mm and BC=49mm. The centre distance between the centres of rotation O and C is 65mm. The path of travel of slider is 11mm below the fixed point C. The slider moves along a horizontal path and OC is vertical.

(4x15=60)
(15)

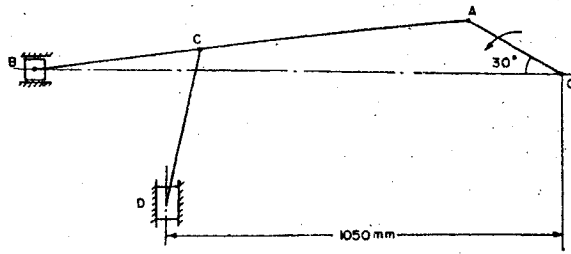
OR

- III. In the mechanism as shown in figure, the crank OA rotates at a uniform speed of 20 rpm anticlockwise and gives motion to the sliding blocks B and D. The dimensions of various links are OA=300mm, AB=1200mm, BC=450mm and CD=450mm for the given configuration. Determine: (i) linear acceleration of D and (ii) angular acceleration of CD.

(15)

(figure overleaf)

(P.T.O)



- IV. a) What is a coupler curve and how the coordinates of a coupler point is obtained? (3)
 b) Explain what is meant by optimum transmission angle? (4)
 c) Derive Freudenstein's equations for four bar and slider crank mechanisms. (8)

OR

- V. For a four bar linkage, the following data are given: (15)

$$\theta = 60^\circ, \phi = 90^\circ$$

$$\frac{d\theta}{dt} = 3 \text{ rad/s}, \frac{d\phi}{dt} = 2 \text{ rad/s}$$

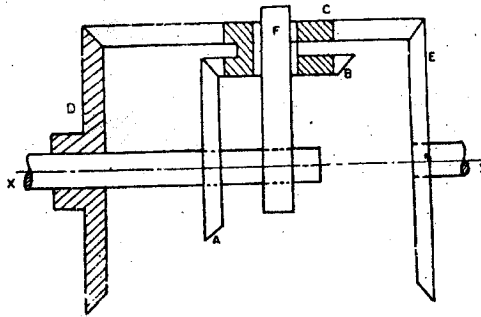
$$\frac{d^2\theta}{dt^2} = -1 \text{ rad/s}^2, \frac{d^2\phi}{dt^2} = 0 \text{ rad/s}^2$$

where ' θ ' is the input link angle and ϕ is the output link angle. Determine the link length ratios.

- VI. Two mating involute spur gears of 20° pressure angle have a gear ratio of 2. The number of teeth on the pinion is 20 and its speed is 250rpm. The module of the teeth is 12mm. If the addendum on each wheel is such that the path of approach and the path of recess on each side are half the maximum possible length each, determine (i) the length of arc of contact and (ii) the maximum velocity of sliding during approach and recess. (15)

OR

- VII. An epicyclic gear train consists of bevel gears as shown in figure. The driving pinion A has 20 teeth and meshes with wheel B which has 25 teeth. The wheels B and C are fixed together and turn freely on a shaft F. The shaft F can freely rotate about the main axis XX. Wheel C has 50 teeth and meshes with wheels D and E, each of which has 60 teeth. Determine the speed and direction of E when A rotates at 200rpm, if D is fixed. (15)



- VIII. Draw the profile of a cam operating a roller reciprocating follower with the following data. (15)

$$\text{Minimum radius of cam} = 25\text{mm}$$

$$\text{Lift} = 30\text{mm}$$

$$\text{Roller diameter} = 15\text{mm}$$

The cam lifts the follower for 120° with SHM followed by a dwell period of 30° . Then the follower lowers down during 150° of the cam rotation with uniform acceleration and deceleration followed by a dwell period. If the cam rotates at a uniform speed of 150 rpm, determine the maximum velocity and acceleration of the follower during ascent and descent.

OR

- IX. The mean diameter of a screw jack having pitch of 12.5mm is 50mm. The coefficient of friction between the screw and the nut is 0.1. Determine the torque required on the screw to raise a load of 30 kN and also the efficiency of the screw jack. (15)